Assignment –
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Ergodicity

1. 1. A random process is defined by $X(t) = \cos(\omega 0t + \Theta)$ where ω_0 is a constant and Θ is random variable uniformly distributed over $(0, \pi)$. Show that the process is mean ergodic or not
2. A random process is defined by $X(t) = 10\cos(100t + \Theta)$where Θ is uniformly distributed over $(-\pi, \pi)$.Show that the process is mean ergodic or not
3. A random process is defined by $X(t) = 10cos\ (100t + \Theta)$ where Θ is a random variable uniformly distributed over $(0, 2\pi)$.
   Show that $X(t)$ is mean and also mean ergodic or not.

4. A random process is defined by $X(t) = Acos(\omega 0t + \Theta)$where $A$and ω_0 are constants and $\Theta$is uniformly distributed on $(-\pi, \pi)$.
   Check whether the process is mean ergodic or not
5. A random process is given by X(t) = Θ for all t where Θ is a uniform random variable in (-1,1). Prove that the process is stationary but not mean ergodic.